
USER MANUAL FOR SEARIBU MARINE INFORMATION SYSTEM: METHODOLOGY AND OPERATIONAL GUIDE

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Preface

This manual is intended to serve as a user guide to the **Searibu** web-based marine information platform. In addition to describing the input and output of all data features on the platform, the manual outlines the scientific methods used in tidal prediction, weather integration, and safety assessment.

Searibu is a capstone design project developed by students of the Geodesy and Geomatics Engineering programme at Institut Teknologi Bandung (ITB), Faculty of Earth Sciences and Technology (FITB). The platform is designed to support marine tourism activities in the Kepulauan Seribu region of Jakarta Bay, Indonesia.

Users who wish to receive updates to the platform and this manual should visit searibu.vercel.app or contact the development team.

Acknowledgements

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Chapter 1

Introduction to Searibu

1.1 Overview

Searibu is an ocean weather-informed marine tourism information system focused on the Kepulauan Seribu (Seribu Islands) National Park area of Jakarta Bay, Indonesia. The platform integrates three primary data sources:

1. **Astronomical tidal prediction** from the TPXO-9 Atlas harmonic model;
2. **Real-time tidal observation** from the automatic tide gauge station operated by Luwes Inovasi Mandiri; and
3. **Marine weather forecast** from the Open-Meteo Marine and Atmospheric APIs.

The system is accessible at searibu.vercel.app and is intended for use by marine tourism operators, recreational visitors, researchers, and maritime professionals operating in the Kepulauan Seribu area.

1.2 Coverage Area

The Searibu platform covers the Kepulauan Seribu (Thousand Islands) archipelago, located approximately 45 – 83 km north of Jakarta in the Java Sea. The spatial extent of the TPXO prediction grid used in the system spans:

Table 1. Spatial coverage of the Searibu system

Parameter	Value	Description
Latitude range	5.3°S – 6.1°S	North-south extent
Longitude range	106.3°E – 107.1°E	East-west extent
Grid resolution	1/30° (~3.7 km)	TPXO spatial resolution
	~9 km (O1280 octahedral grid)	ECMWF IFS HRES spatial resolution
	1/12° (~8 km)	SMOC and MFWAM spatial resolution

1.3 System Architecture

Table 2. Technical stack of the Searibu platform

Component	Technology	Description
Frontend	React / TypeScript / Vite	Web interface, deployed on Vercel
Backend	Python / Flask	API server, deployed on Railway
Database	PostgreSQL	User, subscription, and observation data

Tidal DB	SQLite	Pre-processed TPXO9 harmonic constants
Mapping	Leaflet.js	Interactive WebGIS map
Charts	Chart.js	Tidal and weather visualisation

Chapter 2

Tidal Prediction Methodology

2.1 Harmonic Analysis Background

Astronomical tidal prediction in the Searibu system is computed using the standard harmonic summation formula (Schureman, 1958):

$$h(t) = \sum f_k(t) \cdot A_k \cdot \cos[\omega_k(t - t_0) + V_{0,k}(t_0) + u_k(t) - \kappa_k]$$

where:

- ω_k = angular speed of constituent k (°/hour)
- A_k = amplitude from the TPXO9 database (metres)
- κ_k = Greenwich phase lag (°) from the TPXO9 database
- f_k = nodal amplitude correction factor (Foreman, 1977)
- $V_{0,k}$ = astronomical equilibrium argument at reference epoch t_0
- u_k = nodal phase correction (°)

2.2 TPXO-9 Atlas Model

The system uses the **TPXO-9 Atlas** global barotropic tidal model, developed at Oregon State University by Egbert & Erofeeva (2002). TPXO-9 is a widely recognised model in ocean modelling and hydrographic research for its balance between global coverage and local accuracy.

Table 3. Tidal constituents implemented in the Searibu system

Constituent	Description	Type	Period (h)
M2	Principal lunar semidiurnal	Semidiurnal	12.42
S2	Principal solar semidiurnal	Semidiurnal	12.00
N2	Larger lunar elliptic	Semidiurnal	12.66
K2	Lunisolar semidiurnal	Semidiurnal	11.97
2N2	Lunar elliptical 2nd order	Semidiurnal	12.91
K1	Lunisolar diurnal	Diurnal	23.93
O1	Principal lunar diurnal	Diurnal	25.82
P1	Principal solar diurnal	Diurnal	24.07
Q1	Larger lunar elliptic diurnal	Diurnal	26.87
M4	Shallow water overtide	Shallow	6.21
MS4	Shallow water compound	Shallow	6.10
MN4	Shallow water compound	Shallow	6.27

S1	Solar diurnal	Diurnal	24.00
Mf	Lunisolar fortnightly	Long period	327.9
Mm	Lunar monthly	Long period	661.3

2.3 Vertical Datum

All tidal predictions are referenced to **Mean Sea Level (MSL)**, consistent with IHO S-104 requirements. Heights above MSL are positive; heights below MSL are negative.

Datum Reference Searibu uses MSL (Mean Sea Level) as the vertical datum for all tidal predictions, as defined by IHO S-104. The TPXO-9 Atlas model outputs are already referenced to MSL without additional offset for most locations. A Transfer of Level (TOL) correction of -2.156 m is applied to align observation station raw readings to the MSL datum.

2.4 Transfer of Level (TOL) Correction

The tidal observation station records water levels relative to its sensor reference datum, which differs from MSL. A **Transfer of Level (TOL)** correction is applied:

$$h_{MSL}(t) = h_{observation}(t) + \Delta h_{TOL}$$

where $\Delta h_{TOL} = -2.156$ m is derived from comparison between the field observations and closest benchmark in Pramuka Island.

Chapter 3

Data Sources

3.1 Tidal Observation Station

The tidal observation station is an automatic tide gauge system developed locally and operated by Luwes Inovasi Mandiri. The Searibu system connects to the observation data stream in real time.

Table 4. Tidal observatin station specifications

Parameter	Value
Station name	ITB Pramuka
IMEI	869556066101370
Latitude	5.7439°S
Longitude	106.6128°E
Measurement interval	60 seconds (1 minute)
TOL correction applied	−2.156 m (Luwes → MSL)
Operator	Luwes Inovasi Mandiri

3.2 Open-Meteo Weather and Marine APIs

Marine weather data is sourced from the **Open-Meteo** open-source weather API. Two endpoints are used:

1. **Atmospheric API** (api.open-meteo.com): provides wind speed, wind direction, temperature, and weather codes at 10 m height.
2. **Marine API** (marine-api.open-meteo.com); provides wave height, wave direction, ocean current velocity, and ocean current direction.

All weather data is fetched for a window of 12 days in the past through 5 days into the future, and cached in the browser session to minimise redundant API calls.

Table 5. Weather parameters displayed in the Searibu hourly table

Parameter	Unit	Source	Description
Wind speed	m/s	Atmospheric	10 m above surface
Wind direction	degrees	Atmospheric	Meteorological convention
Temperature	°C	Atmospheric	2 m air temperature
Weather code	WMO code	Atmospheric	Condition category
Wave height	m	Marine	Significant wave height

Wave direction	degrees	Marine	Mean wave propagation dir.
Current speed	m/s	Marine	Surface ocean current
Current dir.	degrees	Marine	Current flow direction

Chapter 4

Using the WebGIS Map

4.1 Map Interface Overview

The WebGIS page is the primary interface for spatially-referenced marine data access. It features:

- **Interactive map** of Kepulauan Seribu;
- **Spatial grid** (blue, orange, and green cells);
- **Island markers** with facility information;
- **Tidal observation station marker** (red hexagon);
- **2 port markers** (Marina Ancol and Muara Angke); and
- **Island search bar** at the bottom.

4.2 Navigating the Map

1. Open the **WebGIS** page from the top navigation bar.
2. The map loads centred on Kepulauan Seribu.
3. Click any grid cell to open the Marine Information Panel on the right (desktop) or as a bottom sheet (mobile).
4. Use the satellite/map toggle (bottom-left) to switch between satellite imagery and OpenStreetMap basemap.
5. Use the island search bar (bottom-centre) to search by island name. Click a result to fly the map to that location.
6. Tap the location button to fly to your current GPS position.

Tip: Grid Selection After clicking a grid cell, it will turn red to indicate it is the active selection. All data in the Marine Information Panel corresponds to the centroid of that cell. Click a different cell at any time to update the panel.

4.3 Map Legend

Table 6. Map symbols used in the Searibu WebGIS

Symbol	Name	Description
Blue circle	Tourism Island	Clickable markers with island details and facilities
Grey pentagon	Port	Ferry departure points
Red hexagon	Tide Station	Luwes automatic tide gauge location
Blue rectangle	TPXO Grid Cell	Click to load tidal and weather data for that location
Orange rectangle	ECMWF Grid Cell	
Green rectangle	SMOC and MFWAM Grid Cell	

Chapter 5

Marine Information Panel

5.1 Panel Overview

Clicking a spatial grid cell opens the **Marine Information Panel**, which consolidates tidal predictions, real-time observations, and weather data for the selected location and date.

5.2 Hero Weather Card

At the top of the panel, a hero card displays:

- **Current air temperature (°C)** and weather condition;
- **Sunrise and sunset times** (WIB, UTC+7);
- **Wind speed (m/s)**, wind direction (cardinal), and wind bearing (°);
- **Wave height (m)** and sea state classification; and
- **Ocean current speed (m/s)** and strength classification.

5.3 Date Selector

Table 7. Date range access by subscription tier

Plan	Historical range	Forecast range
Free Tier	Past 10 days	3 days ahead
Pro Tier	Unlimited	10 days ahead

5.4 Tidal Chart

The tidal chart displays water levels for the selected date from 00:00 to 24:00 WIB. It contains:

- **Blue line:** TPXO-9 Atlas astronomical tidal prediction, smoothed using cubic spline interpolation;
- **Pink dots:** Luwes real-time observations corrected to MSL (when available);
- **Red dashed NOW line:** current WIB time (displayed only for today).

Reading Tidal Heights All heights are in metres above MSL. Positive values indicate water above MSL (high tide); negative values indicate water below MSL (low tide). The TPXO predictions represent astronomical tides only and do not include storm surges or meteorological effects.

5.5 Hourly Data Table

Table 8. Columns in the hourly data table

Column	Unit	Source	Notes
Time	WIB (HH:MM)	—	00:00 to 23:00
Wx	WMO code	Open-Meteo	Weather symbol
Tide	m (MSL)	TPXO9	Highlighted at max (blue) and min (amber)
Temp	°C	Open-Meteo	2 m air temperature
Wind	m/s + dir.	Open-Meteo	Speed and 8-point cardinal direction + arrow
Wave	M + dir	Open-Meteo Marine	Significant wave height + direction
Curr.	m/s + dir.	Open-Meteo Marine	Surface current speed + direction

Table 9. Colour coding for wave height and current speed

Colour	Wave (m)	Current (m/s)	Interpretation
Green	< 0.50	< 0.25	Low / calm
Amber	0.50–1.25	0.25–0.75	Moderate
Red	> 1.25	> 0.75	High / strong

Chapter 6

Activity Safety Assessment

6.1 Overview

The **Activity Safety Assessment** is available both on the Home page (as a standalone safety check tool) and within the Marine Information Panel (based on the selected grid cell and date). It evaluates conditions for 11 marine tourism activities using a three-tier status system.

6.2 Safety Status Levels

Table 10. Activity safety status definitions

Status	Definition
Safe	Conditions are within safe operational limits. Suitable for all participants.
Caution	Moderate conditions. Exercise caution; recommended for experienced practitioners only.
Avoid	Conditions exceed safe operational limits. Do not attempt this activity.

6.3 Activity Thresholds

Safety thresholds are based on Chuang et al. (2024) and de Vos & Rautenbach (2019). All three parameters must pass for an activity to receive Safe status.

Table 11. Safety thresholds for marine activities in the Searibu system

Activity	Wave Safe (m)	Wave Caution (m)	Wind Safe (m/s)	Wind Caution (m/s)
Snorkeling	< 0.50	0.50–1.00	< 3.3	3.3–7.9
Scuba Diving	< 0.50	0.50–1.25	—	—
Freediving	< 0.50	0.50–0.80	—	—
Jet Ski	< 0.80	0.80–1.50	< 7.9	7.9–10.3
SUP / Kayak	< 0.50	0.50–1.00	< 4.5	4.5–6.2
Island Hopping	< 1.00	1.00–1.50	< 7.9	7.9–10.3
Fishing	< 1.00	1.00–1.50	< 7.9	7.9–10.3
Turtle Watch	any	any	any	any
Beach Camping	any	any	any	any
UW Photography	< 0.50	0.50–1.00	—	—

General Tourism	any	any	any	any
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Note Note: Turtle Watch, Beach Camping, and General Tourism are primarily affected by weather code (WMO). If a thunderstorm code (≥ 95) is detected, the activity receives Avoid status. Rain codes (≥ 51) trigger Caution status.

6.4 Using the Home Page Safety Check

1. On the Home page, scroll to the Activity Safety Check section.
2. Select an island from the dropdown list (12 islands available).
3. Choose a date using the date picker (range: past 10 days to 3 days ahead).
4. Click Analyse Conditions.
5. The system fetches live weather data and computes the safety rating for all 11 activities.
6. A summary shows the count of Safe / Caution / Avoid activities.
7. Average condition metrics (wind, wave height, current, weather) are displayed for the selected date.

Chapter 7

IHO S-104 Compliance

7.1 Standards Overview

The Searibu system implements two international hydrographic data standards:

- **IHO S-100** — Universal Hydrographic Data Model; and
- **IHO S-104** — Water Level Information for Surface Navigation.

S-104 is a product specification within the S-100 framework, specifically designed for encoding tidal and water level data for use in ECDIS (Electronic Chart Display and Information System) environments.

7.2 S-104 Implementation Details

Table 12. S-104 implementation attributes in the Searibu system

Attribute	Value	Description
productSpecification	INT.IHO.S-104.2.0	S-104 Edition 2.0.0
horizontalCRS	4326	WGS 84 (EPSG:4326)
verticalCoordinateBase	2	verticalDatum (mandatory Ed.2)
verticalDatum	12	Mean Sea Level (MSL)
dataDynamicity (TPXO)	1	astronomicalPrediction
dataDynamicity (Observation)	3	observed
dataCodingFormat	1	Time series at fixed stations
timeRecordInterval	3600 s	1-hour interval (TPXO)
TOL Correction	-2.156 m	Luwes → MSL TPXO9
Encoding	HDF5	Hierarchical Data Format v5

7.3 Exporting S-104 HDF5 Files

1. Open the **WebGIS** page and click a grid cell to open the Information Panel.
2. Scroll to the IHO S-104 section at the bottom of the panel.
3. Click Export S-104 (TPXO) to download a 24-hour astronomical prediction as an HDF5 file for the selected date.
4. Click Export S-104 (Luwes) to download real observations for the selected date as an HDF5 file.

The resulting .h5 files are fully compliant with IHO S-104 and can be opened using:

- **HDFView** (free, MATLAB, or HDF Group tool);
- s100py Python library;
- Any ECDIS supporting IHO S-100/S-104.

Chapter 8

Account and Subscription

8.1 Subscription Plans

Table 13. Searibu subscription plan comparison

Feature	Free	Pro Monthly	Pro Annual
Price (IDR)	Rp 0	Rp 39.000/bln	Rp 139.000/thn
Tidal forecast range	3 days ahead	14 days ahead	14 days ahead
Historical data range	Past 10 days	Past 14 days	Past 14 days
S-104 export	No	Yes	Yes
Activity guide	Past 10 days and 3 days ahead	All dates	All dates
Hourly weather table	Yes	Yes	Yes

8.2 Registering an Account

1. Click **Sign In** in the top navigation bar, then select Sign up.
2. Enter your full name, email address, and a password of at least 6 characters. Click Continue.
3. On Step 2, select your account type: General User or Researcher / Professional.
4. Click **Create Account**. You will be logged in immediately.
5. Alternatively, use Continue with Google to sign in with a Google account.

8.3 Upgrading to Pro

1. Click the **Free** badge in the top navigation bar (or click your name and open the Profile menu).
2. The Pricing modal opens. Select Pro Monthly or Pro Annual.
3. Click **Upgrade**, then complete payment via Midtrans (supports credit card, bank transfer, and e-wallet).
4. Your subscription activates immediately after payment confirmation.

Chapter 10

Islands of Kepulauan Seribu

Table 14. Islands available in the Searibu system

Island	Latitude	Longitude	Key Features
Pulau Bidadari	6.035°S	106.746°E	VOC fortress, resort, snorkeling
Pulau Tidung	5.797°S	106.497°E	Love Bridge, cycling, snorkeling
Pulau Pari	5.858°S	106.618°E	Starfish, seagrass, LIPI research
Pulau Kelapa	5.654°S	106.569°E	Diving, snorkeling, speedboat
Pulau Pramuka	5.745°S	106.614°E	Administrative centre, resort
Pulau Untung Jawa	5.977°S	106.706°E	Day trips, white sand beach
Pulau Kotok	5.701°S	106.539°E	Turtle conservation, world-class diving
Pulau Putri	5.594°S	106.560°E	Underwater observatory, premium resort
Pulau Ayer	5.764°S	106.583°E	Overwater bungalows, seafood
Pulau Rambut	5.975°S	106.692°E	Seabird sanctuary, mangrove
Pulau Lancang	5.930°S	106.587°E	Fishing village, calm beaches
Pulau Bokor	5.978°S	106.707°E	Uninhabited, picnic, snorkeling

Chapter 11

Frequently Asked Questions

Q: Why does the tidal chart show negative heights?

A: Negative values indicate water levels below Mean Sea Level (MSL). This is normal during low tide conditions. The datum is MSL, not the seabed.

Q: Why are there no orange dots (observations) on the chart?

A: Luwes data may not be available for the selected date due to station maintenance or data gaps. The TPXO prediction line (blue) is always available for any date within the accessible range.

Q: Why is my wave height data the same across different grid cells?

A: Open-Meteo Marine API has a resolution of approximately 8 km. For closely spaced grid cells within one Open-Meteo grid box, the wave data may be identical.

Q: What does the NOW line on the tidal chart represent?

A: The red dashed line indicates the current WIB (UTC+7) time. It appears only when today's date is selected.

Q: Can I download historical tidal data?

A: Yes, via the IHO S-104 HDF5 export feature. The exported file contains all hourly predictions for the selected date in a standards-compliant HDF5 file.

Appendix A: API Endpoint Reference

Table 15. Searibu backend API endpoints

Method	Endpoint	Description
GET	/api/tide/prediction	TPXO9 tidal prediction for a location and date range
GET	/api/luwes/level	Latest Luwes water level observation
GET	/api/luwes/overlay	Combined Luwes + TPXO data for a date
GET	/api/s104/export	Download S-104 HDF5 (TPXO predictions)
GET	/api/s104/export/luwes	Download S-104 HDF5 (Luwes observations)
GET	/api/s104/metadata	S-100/S-104 compliance metadata
GET	/api/subscription	User subscription status
POST	/api/auth/register	Register a new account
POST	/api/auth/login	Email + password authentication
POST	/api/auth/google	Google OAuth sign-in
PUT	/api/profile/role	Update user account type
GET	/api/admin/stats	Platform statistics (admin only)
GET	/api/admin/payments	Recent payment list (admin only)

Appendix B: WMO Weather Code Reference

Table 16. WMO weather codes displayed in the Searibu hourly table

Code	Condition (EN)	Code	Condition (ID)
0	Clear sky	0	Langit cerah
1	Mainly clear	1	Cerah berawan
2	Partly cloudy	2	Sebagian berawan
3	Overcast	3	Mendung
45	Foggy	45	Berkabut
51	Light drizzle	51	Gerimis ringan
61	Light rain	61	Hujan ringan
63	Moderate rain	63	Hujan sedang

65	Heavy rain	65	Hujan lebat
80	Light showers	80	Hujan rintik
95	Thunderstorm	95	Badai petir